Final Research Output

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```
import numpy as np
import pandas as pd

def load_npz(path):
    d = np.load(path, allow_pickle=True)
    return pd.DataFrame(d["data"], columns=d["columns"], index=d["index"])

hv_a = load_npz("/Users/useradmin/Desktop/Files/ivol_OEX_historical_volatility_2016-01-01_to_2019-12-31
hv_b = load_npz("/Users/useradmin/Desktop/Files/ivol_OEX_historical_volatility_2020-01-01_to_2026-09-30
rates = load_npz("/Users/useradmin/Desktop/Files/ivol_interest_rates_USD_2016-01-01_to_2026-09-30.npz")

# Volatility: combine, parse dates, convert only the numeric columns
hv = pd.concat([hv_a, hv_b], ignore_index=True)
hv["date"] = pd.to_datetime(hv["date"])
num_cols = ["price"] + [c for c in hv.columns if c.endswith("HV")]
hv[num_cols] = hv[num_cols].astype(float)
hv = hv[["date"] + num_cols].sort_values("date").set_index("date")

# Rates
rates["t_date"] = pd.to_datetime(rates["t_date"])
rates["period"] = rates["period"].astype(int)
rates["rate"] = rates["rate"].astype(float)

print(hv.head())
```

```
##              price    10d HV    20d HV  ...   120d HV   150d HV   180d HV
## date                                    ...
## 2016-01-04  896.54  0.180150  0.192799  ...  0.190700  0.178967  0.168660
## 2016-01-05  897.66  0.150883  0.174295  ...  0.190705  0.178974  0.168229
## 2016-01-06  886.78  0.156092  0.177415  ...  0.191014  0.179589  0.168815
## 2016-01-07  864.48  0.184473  0.195153  ...  0.194266  0.182217  0.171263
## 2016-01-08  855.58  0.160943  0.196274  ...  0.194706  0.182652  0.171649
##
## [5 rows x 9 columns]
```

```
print(hv.tail())
```

```
##               price    10d HV    20d HV  ...   120d HV   150d HV   180d HV
## date                                     ...
## 2026-09-24  3847.68  0.131835  0.117045  ...  0.138137  0.146449  0.143669
## 2026-09-25  3868.18  0.128577  0.113727  ...  0.138178  0.146325  0.143772
## 2026-09-28  3836.54  0.134827  0.118088  ...  0.138895  0.146148  0.144143
## 2026-09-29  3825.43  0.133375  0.118085  ...  0.135087  0.146010  0.144113
```

```
## 2026-09-30  3822.99  0.130431  0.115956  ...  0.134729  0.145595  0.144117
##
## [5 rows x 9 columns]
```

```
print(rates.head())
```

```
##       t_date  period currency    rate
## 0 2016-01-04       1      USD  0.4295
## 1 2016-01-04       2      USD  0.4295
## 2 2016-01-04       3      USD  0.4295
## 3 2016-01-04       4      USD  0.4295
## 4 2016-01-04       5      USD  0.4295
```

```
knitr::kable(head(py$hv, 5), digits = 4, caption = "OEX historical volatility (first 5 rows)")
```

Table 1: OEX historical volatility (first 5 rows)

	price	10d HV	20d HV	30d HV	60d HV	90d HV	120d HV	150d HV	180d HV
2016-01-03 18:00:00	896.54	0.1802	0.1928	0.1680	0.1544	0.1890	0.1907	0.1790	0.1687
2016-01-04 18:00:00	897.66	0.1509	0.1743	0.1682	0.1537	0.1759	0.1907	0.1790	0.1682
2016-01-05 18:00:00	886.78	0.1561	0.1774	0.1706	0.1557	0.1726	0.1910	0.1796	0.1688
2016-01-06 18:00:00	864.48	0.1845	0.1952	0.1840	0.1642	0.1779	0.1943	0.1822	0.1713
2016-01-07 18:00:00	855.58	0.1609	0.1963	0.1851	0.1651	0.1782	0.1947	0.1827	0.1716

```
knitr::kable(tail(py$hv, 5), digits = 4, caption = "OEX historical volatility (last 5 rows)")
```

Table 2: OEX historical volatility (last 5 rows)

	price	10d HV	20d HV	30d HV	60d HV	90d HV	120d HV	150d HV	180d HV
2026-09-23 19:00:00	3847.68	0.1318	0.1170	0.1061	0.1227	0.1375	0.1381	0.1464	0.1437
2026-09-24 19:00:00	3868.18	0.1286	0.1137	0.1056	0.1230	0.1377	0.1382	0.1463	0.1438
2026-09-27 19:00:00	3836.54	0.1348	0.1181	0.1080	0.1244	0.1379	0.1389	0.1461	0.1441
2026-09-28 19:00:00	3825.43	0.1334	0.1181	0.1069	0.1236	0.1369	0.1351	0.1460	0.1441
2026-09-29 19:00:00	3822.99	0.1304	0.1160	0.1059	0.1234	0.1369	0.1347	0.1456	0.1441

```
knitr::kable(head(py$rates, 5), digits = 4, caption = "USD interest rates (first 5 rows)")
```

Table 3: USD interest rates (first 5 rows)

	t_date	period	currency	rate
0	2016-01-03 18:00:00	1	USD	0.4295
1	2016-01-03 18:00:00	2	USD	0.4295
2	2016-01-03 18:00:00	3	USD	0.4295
3	2016-01-03 18:00:00	4	USD	0.4295
4	2016-01-03 18:00:00	5	USD	0.4295